

Lab - 8

Generate the Frequent Itemset Tree for the given dataset.

1	TID	Itemset	Minimum Support = 3
	1	E K M N O Y	
	2	D E K N O Y	
	3	A E K M	
	4	C K M U Y	
	5	C E I K O	

Step - 1

Item	Frequency
A	1
BC	2
D	1
E	4
I	1
K	5
M	3
N	2
O	3
U	1
Y	3

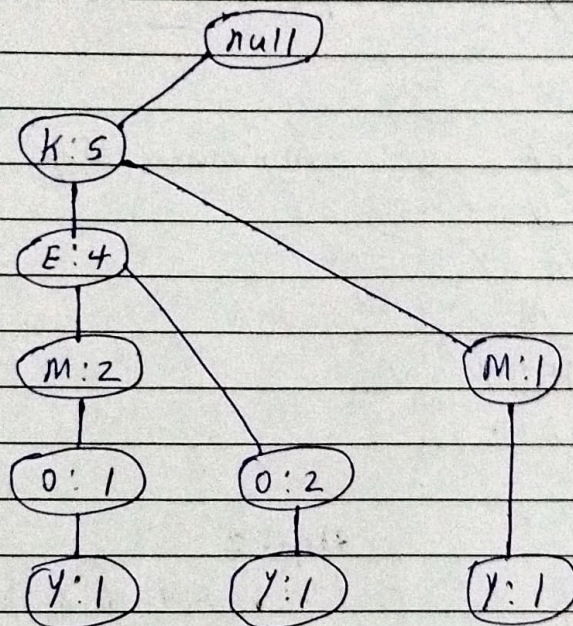
Step - 2

{ K:5, E:4, O:3, M:3, Y:3 }

TID	Sorted Items
1	K E M O Y
2	K E O Y
3	K E M
4	K M Y
5	K E O

Step - 3

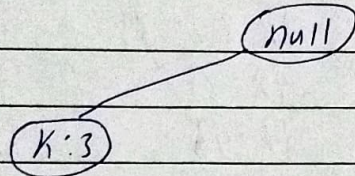
Building the FP-Tree



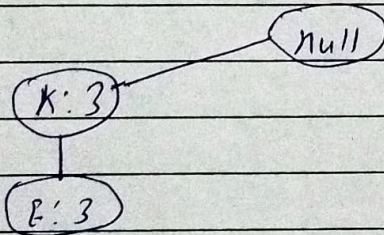
Conditional Pattern Base:-

Item	Condition Pattern Base
Y	{KEMO:1} {KEO:1} {KM:1}
O	{KEM:2} {KE:2}
M	{KE:2} {K:1}
E	{K:4}
K	-

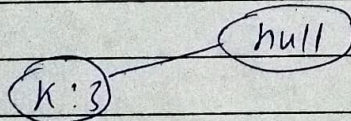
Item	Conditional Pattern Base	Conditional FP base
Y	{KEmo:1} {KGO:1} {KM:1}	{K:3}



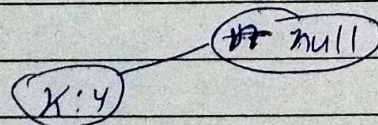
Item	Conditional Pattern Base	Conditional FP base
O	{KEom:1} {KE:2}	{K:3} {E:3}



Item	Conditional Pattern Base	Conditional FP base
M	{KE:2} {K:1}	{K:3}



Item	Conditional Pattern Base	Conditional FP base
E	{K:4}	{K:4}



Item	Conditional Pattern Base	Conditional FP Tree	Frequent Pattern Generated
Y	$\{XEMO:1\} \{XEO:1\}$	$\{X:3\}$	$\{X,Y:3\}$
0	$\{XAM:1\}$		X
O	$\{XEM:1\} \{XB:2\}$	$\{X:3, E:3\}$	$\{X,O:3\} \{EO:3\} \{XEO:3\}$
M	$\{XB:2\} \{X:1\}$	$\{X:3\}$	$\{X,m:3\}$
E	$\{X:4\}$	$\{X:4\}$	$\{X,E:4\}$
X	—	—	—

2

2.	TID	Items	Minimum Support = 2
	1	1 2 5	
	2	2 4	
	3	2 3	
	4	1 2 4	
	5	1 3	
	6	2 3	
	7	1 3	
	8	1 2 3 5	
	9	1 2 3	

Step-1

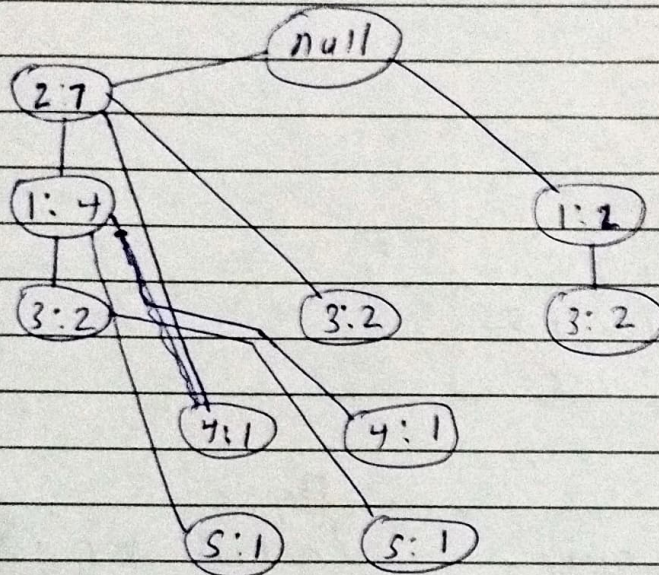
Item	Frequency
1	6
2	7
3	6
4	2
5	2

Step-2

{ 2: 7, 1: 6, 3: 6, 4: 2, 5: 2 }

TID	Items
1	2 1 5
2	2 4
3	2 3
4	2 1 4
5	1 3
6	2 3
7	1 3
8	1 3 5
9	2 1 3

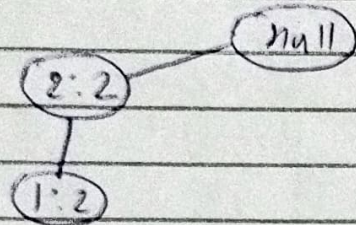
Step - 3
Building the FP-Tree



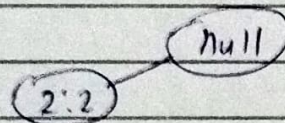
Conditional Pattern Base:-

Item	Conditional Pattern Base
5	{2,1:1} {2,1,3:1}
4	{2:1} {2,1:1}
3	{2,1:2} {2:2} {1,2}
1	{2:4}
2	-

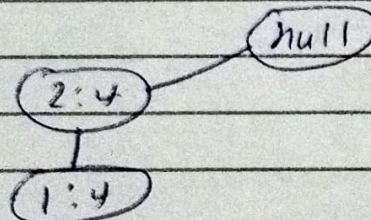
Item	Conditional Pattern Base	Conditional FP base
5	$\{2, 1:1\} \{2, 1, 3:1\}$	$\{2:2, 1:2\}$



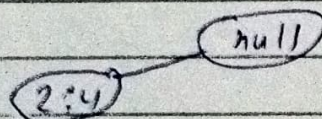
Item	Conditional Pattern Base	Conditional FP base
4	$\{2:1\}, \{2, 1:1\}$	$\{2:2\}$



Item	Conditional Pattern Base	Conditional FP base
3	$\{2, 1:2\} \{2:2\} \{1:2\}$	$\{2:4, 1:2\}$



Item	Conditional Pattern Base	Conditional FP tree
1	$\{2:4\}$	$\{2:4\}$



Item	Conditional Pattern	Conditional FP Tree	Fragment Pattern Generated
5	{2,1:1} {2,1,3:1}	{2:2}, {1:2}	{2,5:2}, {1,5:2} {2,1,5:2}
4	{2:1} {2,1:1}	{2:2}	{2,4:2}
3	{2,1:2}, {2:2} {1:2}	{2:4}, {1:4}	{2,3:4}, {2,1:4} {2,1,3:4}
2	{2:4}	{2:4}	{2,2:4}
1	-	-	-